

State of the Art in Quantum Convolutional Neural Networks for Image Classification

Mariel Álvarez Salas

1 Introduction

Quantum convolutional neural networks (QCNNs) are quantum neural network (QNN) architectures inspired by the hierarchical structure of classical convolutional neural networks (CNNs), although the term convolution is used differently across the literature. In the foundational QCNN model proposed by [Cong et al. \(2019\)](#), convolutional layers apply quasi-local unitary operations in a translation-invariant manner, while pooling layers reduce the number of active qubits and degrees of freedom through partial measurements and unitary operations conditioned on their outcomes.

In image-classification research, however, the term QCNN is often used more broadly to describe architectures that either implement quantum analogues of convolution and pooling or integrate parameterized quantum circuit (PQC) blocks into classical image-processing pipelines ([Wei et al., 2022](#); [Henderson et al., 2020](#)).

The literature therefore shows that QCNN is not a single fixed architecture, but rather a family of hierarchical quantum models whose structure depends on the input representation, feature-encoding strategy, pooling mechanism, circuit ansatz, and degree of hybridization with classical neural networks ([Hur et al., 2022](#)).

Applying QCNNs to image classification introduces challenges that are largely absent from the original quantum-data setting considered by [Cong et al. \(2019\)](#). Classical images must first be encoded into quantum states, often requiring dimensionality reduction, feature compression, or patch-based processing to fit within the limited qubit budgets available on current noisy intermediate-scale quantum (NISQ) devices. As a consequence, image-oriented QCNN research has evolved along several directions, including the development of more efficient encoding strategies, trainable quantum convolutional filters, alternative pooling mechanisms, hybrid quantum–classical architectures, and resource efficient circuit designs.

The resulting landscape is highly heterogeneous, with substantial variation in datasets, qubit counts, optimization procedures, evaluation methodologies, and hardware assumptions. Understanding these design choices and their impact on performance is therefore essential for assessing the current state of the field.

Given the rapidly expanding and increasingly diverse body of work, a systematic synthesis of recent developments is needed. This review analyzes representative state-of-the-art QCNN models for image classification, focusing on the architectural and methodological choices that characterize modern QCNN research. By identifying common design patterns, emerging trends, and persistent limitations, this review aims to provide a comprehensive overview of the current state of QCNN-based image classification and to highlight promising directions for future research.

2 State-of-the-Art QCNN Models for Image Classification

Recent work on quantum convolutional neural networks has explored different strategies for adapting convolutional architectures to quantum image classification. These approaches vary considerably in the way classical images are encoded into quantum states, the design of quantum convolution and pooling operations, the number of qubits required, and whether the models are evaluated on simulators or quantum hardware.

Table 1 summarizes representative state-of-the-art QCNN and quanvolutional models for image classification, highlighting the datasets used, reported performance, computational backend, qubit requirements, and the main methodological contribution of each work.

Table 1: State-of-the-art QCNN models for image classification.

Paper	Year	Dataset	Results	Qubits	Backend	Key Contribution
(Henderson et al., 2020)	2020	MNIST	>95% test accuracy	9	QxBranch Quantum Computer Simulation System	Introduced the quanvolutional layer: fixed random quantum circuits used as local feature extractors within a CNN.
(Li et al., 2020)	2020	MNIST; GTSRB	98.97%; 91.40% test accuracy	$2n + 5d + p + 2$	Qiskit	Introduces a QDCNN based on parameterized quantum circuits, QRAM-based basis encoding, QPE-based nonlinear mapping, and hybrid quantum-classical parameter training. Pooling layer is omitted and subsampling is implemented through increased convolutional stride.
(Wei et al., 2022)	2022	MNIST (2 classes: 1 vs 8); (10 classes)	96.3% noise-free and 96.0% noisy test accuracy; 74.3% noise-free and 74.0% noisy test accuracy	10 work + 4 ancilla	-	The QCNN realizes convolution through linear combinations of unitary operators, pooling by ignoring selected work-register qubits, and classification via expectation values of parameterized Hamiltonians. In experiments, noisy circuits were modeled using a stochastic single-qubit Pauli channel, applying a randomly selected Pauli error after each operation with a fixed probability.
(Hur et al., 2022)	2022	MNIST (2 classes: 1 vs 0); Fashion MNIST (2 classes: 1 vs 0)	$98.7 \pm 0.1\%$; $94.3 \pm 1.2\%$ respectively for best performing ansatz	8	PennyLane	Benchmark using literature-inspired convolution ansätze. Best MNIST result used Ansatz 8, chosen for strong expressibility/entangling capability, with Dense encoding + PCA. Best Fashion-MNIST result used Ansatz 6, a VQE-inspired two-body entangler/SO(4) candidate, with HDE + AutoEncoder.
(Bokhan et al., 2022)	2022	MNIST (4 classes: 3/4/5/6 and 0/1/2/3); Fashion-MNIST (4 classes: 0/1/2/3 and 1/2/8/9)	85.14% and 90.03%; 85.93% and 93.63%	8 data qubits + 4 ancilla	Cirq, TensorFlow Quantum	Proposes a hybrid quantum-classical QCNN with a preliminary trainable n-qubit scanning layer, introducing parameterized entangling filters that learn flexible quantum feature maps before pooling and softmax readout.

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Table 1: State-of-the-art QCNN models for image classification continued.

Paper	Year	Dataset	Results	Qubits	Backend	Key Contribution
(Shi et al., 2024)	2024	MNIST (4-classes: 0–3 and 3–6)	91.60% and 89.22% for best performing model and configuration	8	PennyLane; TensorFlow	Proposes NISQ-compatible QCNN architectures (QCNN-NN, QCNN-CB, QCNN-AA) by varying the sliding mode/topology of two-qubit entangling quantum filters. Uses dense qubit coding on 24 PCA features into 8 qubits and compares five PQC filter designs plus shared vs unshared filter parameters. Best model resulted in U_5 filter and removing parameter sharing with QCNN-AA.
(Song et al., 2024)	2024	MNIST; Fashion-MNIST	MNIST: high multiclass performance reported via ROC; Fashion-MNIST: 92.94%	5 per filter	-	Proposes RE-QCNN using QAOA-based quantum filters with parameter sharing. The model uses two quantum convolutional layers with pooling and a classical fully connected layer. The experiments showed robustness to measurement errors, and improved performance with deeper QAOA circuits.
(Mordacci et al., 2024)	2024	MNIST (multi-class: 0–3, 0–5, 0–7, and 0–9)	85% / 72% / 70% / 57% accuracy	8	PennyLane	Multi-class QCNN using PCA + amplitude encoding, preconvolutional expressibility filters, $SU(4)$ two-qubit convolutional filters, and pooling.
(Röseler et al., 2025)	2025	MNIST (2 classes: 0 vs 1)	96.08% accuracy	49	Qiskit, IBM Heron r2 quantum processor	Introduces fragment encoding to process full 28×28 MNIST images without PCA or resizing, plus an ansatz-search framework for compact QCNN convolution circuits based on expressibility, entanglement, and circuit complexity.
(Sun et al., 2025)	2025	MNIST (2 classes: 0 vs 1, 2 vs 5, 3 vs 6); Fashion-MNIST (2 classes: 2 vs 5, 4 vs 7, 8 vs 9)	97.9% average accuracy; 99.79% average accuracy	16, 64	TensorFlow Quantum simulation	Proposes SQCNN, a scalable multi-device QCNN where image segments are processed in parallel by independent quantum feature extractors, then aggregated by a final quantum prediction layer. Experiments on 16-qubit version and a 64-qubit version, the last one achieved best results.
(Prajapat et al., 2025)	2025	MNIST (2 classes: 0 vs 1); Wine class 0 vs. not class 0	397.26% test accuracy; 97.22% test accuracy	8	PennyLane	Proposes a compact 45-parameter QCNN for NISQ-compatible binary classification, using 8-qubit R_y angle encoding and three trainable quantum convolutional layers based on two-qubit U_{SU} blocks with R_y , R_z , $U3$, and CNOT gates; combines local, long-range, and final feature-aggregation interactions before single-qubit Pauli-Z measurement.
(Daka and Bhat-tacharyya, 2026)	2026	Synthetic low-resolution colour images: 4×4 binary line dataset; 8×8 six-class RGB line dataset	89.7% binary validation accuracy; 82.05% six-class validation accuracy on 50-image datasets;	16	IBM Qiskit local simulator; Qiskit Machine Learning VQC	Introduces No-QCNN, an end-to-end quantum convolution–pooling architecture for pixelated RGB images, using a spatial–chromatic block-matrix encoding with ZZFeatureMap, two-qubit XXXZ-inspired quantum convolution units, quantum pooling for qubit downsampling, and COBYLA-optimized VQC training.

The early development of QCNNs for image classification is characterized by the use of quantum circuits as mechanisms for local feature extraction. At this stage, the models are primarily aimed

at demonstrating the feasibility of replacing or complementing classical convolutional layers with quantum circuits, either through fixed quantum convolutional layers or through parameterized QDCNN architectures with quantum encoding and hybrid training (Henderson et al., 2020; Li et al., 2020).

Starting in 2022, the field is moving toward more structured and trainable QCNN architectures. Instead of using quantum circuits solely as fixed transformations, research is beginning to compare ansatzes, encoding mechanisms, pooling strategies, and entanglement structures for binary and reduced multiclass classification (Wei et al., 2022; Hur et al., 2022; Bokhan et al., 2022). This stage marks a transition from exploratory demonstrations toward a more systematic design of quantum filters and parameterized convolutional layers.

In 2024, research shifted toward the explicit exploration of the QCNN architectural space. Studies from this period analyze how interaction topology, the use of shared parameters, filter expressiveness, and resource efficiency affect model performance (Shi et al., 2024; Song et al., 2024; Mordacci et al., 2024). In particular, it is observed that the quantum filter architecture becomes a central component of the design, not merely an auxiliary block within a hybrid network.

The most recent works show a growing concern for scalability, execution on real hardware, and reducing dependence on classical preprocessing. In this vein, strategies such as fragment encoding for processing full images, multi-device architectures for parallelizing feature extraction, compact NISQ-compatible circuits, and specialized designs for low-resolution RGB images have emerged (Röseler et al., 2025; Sun et al., 2025; Prajapat et al., 2025; Daka and Bhattacharyya, 2026). This stage suggests that current research is attempting to move QCNNs from reduced benchmarks toward configurations that more closely resemble practical hardware and visual data constraints. It is important to highlight that MNIST and Fashion-MNIST remain the predominant benchmarks, typically in binary or few-class versions. Although GTSRB, Wine, synthetic RGB images, and Overhead-MNIST have emerged, most results are still based on controlled scenarios. This suggests that the field is still in a phase of architectural validation rather than a stage of deployment on complex and realistic visual datasets.

2.1 Encoding Strategies and Preprocessing

Data encoding is consistently identified as one of the central bottlenecks when applying QCNNs to classical image data. Since image pixels are classical, the input must first be compressed, normalized, or mapped into quantum states before quantum convolutional layers can operate.

Early quantum convolutional approaches avoid a full quantum-image representation by applying fixed random quantum circuits to local image patches, using the quantum circuit mainly as a local feature transformation inside an otherwise classical pipeline (Henderson et al., 2020). In contrast, quantum-image-processing-oriented QCNNs encode pixel positions into computational basis states and pixel intensities into amplitudes, enabling convolutional operations over an amplitude-encoded image representation (Wei et al., 2022).

A systematic comparison of encoding and preprocessing choices is provided by Hur et al. (2022), who evaluate amplitude encoding, qubit encoding, dense encoding, hybrid direct encoding, and hybrid angle encoding in combination with bilinear interpolation, PCA, and autoencoding. Their results show that no encoding strategy is universally optimal: PCA performs better on average for MNIST in several reported settings, whereas autoencoding is more competitive for Fashion-MNIST. This indicates that QCNN performance is strongly data-dependent and that preprocessing cannot be treated as a neutral implementation detail.

Recent work increasingly attempts to reduce the encoding bottleneck either by compressing features more efficiently or by designing task-specific encodings. Shi et al. (2024) use dense qubit coding based on H - R_z - R_y - R_z gates, allowing three classical values to be encoded per qubit and

reducing qubit requirements relative to standard angle encoding. [Mordacci et al. \(2024\)](#) compare amplitude and angle encoding for multiclass MNIST and report that angle encoding deteriorates more rapidly as the number of classes increases, whereas amplitude encoding is more robust in their multiclass setting. [Song et al. \(2024\)](#) adopt amplitude encoding for sparse image blocks in a QAOA-based convolutional layer, arguing that sparse state preparation can reduce the loading cost to $O(k \log N)$ under sparsity assumptions. However, this advantage depends on the sparsity of the data and should not be generalized to dense images.

Other recent architectures prioritize implementation practicality over asymptotic compactness. [Prajapat et al. \(2025\)](#) use R_y angle encoding for binary classification, favoring a simple NISQ-compatible encoding with shallow rotations. Similarly, [Sun et al. \(2025\)](#) and [Liu and Lou \(2026\)](#) employ angle encoding in scalable or hybrid QCNN designs, reflecting the broader trend that angle-based encodings remain attractive because they are straightforward to implement, even though they require one qubit per input feature. [Röseler et al. \(2025\)](#) address this limitation through fragment encoding, which processes full 28×28 MNIST images without PCA or resizing by splitting the image into smaller fragments. Finally, [Daka and Bhattacharyya \(2026\)](#) extend encoding design to low-resolution RGB images by using a spatial–chromatic block-matrix representation with ZZFeatureMap, explicitly preserving both pixel position and colour information before quantum convolution and pooling.

In general, the literature shows a clear trade-off. Table 2 synthesizes the encoding and preprocessing approaches reported across the reviewed QCNN studies, highlighting the trade-offs between qubit efficiency, implementation complexity, scalability, and information preservation.

Table 2: Encoding and preprocessing strategies reported in the QCNN literature.

Strategy	Reported use	Implication
Amplitude encoding	Used in quantum-image QCNNs and resource-efficient QCNNs (Wei et al., 2022; Song et al., 2024; Mordacci et al., 2024)	Qubit-efficient and useful for compact image representations; efficient state preparation is mainly justified under sparsity or compression assumptions, but dense state preparation can be costly.
Angle / qubit-style encoding	Used or evaluated in several NISQ-compatible QCNNs (Hur et al., 2022; Prajapat et al., 2025; Sun et al., 2025; Liu and Lou, 2026)	Simple and hardware-friendly, but typically requires one qubit per input feature, making dimensionality reduction or patch-based processing necessary.
Dense encoding	Used by Shi et al. (2024)	Encodes multiple classical values per qubit using rotation-gate sequences, reducing qubit overhead compared with standard angle encoding.
Hybrid direct and hybrid angle encoding	Evaluated by Hur et al. (2022)	Explores trade-offs between circuit width, depth, normalization, and performance; results indicate that encoding quality depends strongly on dataset and preprocessing.
Fragment encoding	Proposed by Röseler et al. (2025)	Processes full MNIST images without PCA or resizing by fragmenting the input, reducing dependence on classical dimensionality reduction.
Spatial-chromatic ZZFeatureMap encoding	Proposed by Daka and Bhat-tacharyya (2026)	Encodes both RGB colour and spatial position for low-resolution colour images, preserving spatial-chromatic correlations before quantum convolution and pooling.
Classical preprocessing: PCA, autoencoding, resizing, patching	Used across multiple QCNN pipelines (Hur et al., 2022; Shi et al., 2024; Mordacci et al., 2024)	Remains necessary in many models because current quantum circuits cannot directly process high-dimensional images at scale; preprocessing choice can substantially affect accuracy.

2.2 Circuit Structures, Qubits, Parameters, and Evaluation Metrics

Although QCNN architectures differ substantially in their convolutional filters, pooling mechanisms, and feature aggregation strategies, most reported models remain relatively small due to NISQ hardware constraints. The majority of image-classification QCNNs reviewed in this survey operate with fewer than ten data qubits, typically using 8-qubit architectures (Hur et al., 2022; Bokhan et al., 2022; Shi et al., 2024; Mordacci et al., 2024; Prajapat et al., 2025), while specialized designs employ 4- or 5-qubit quantum filters embedded within larger hybrid architectures (Song et al., 2024; Liu and Lou, 2026). Only recent scalability-oriented approaches substantially increase the

quantum footprint, such as the 16- and 64-qubit multi-device SQCNN architecture proposed by [Sun et al. \(2025\)](#).

A recurring design objective is parameter efficiency. Early QCNN-inspired models emphasized logarithmic parameter scaling, while later architectures explored the trade-off between expressibility and trainability through increasingly diverse circuit designs. For example, [Shi et al. \(2024\)](#) systematically vary both filter topology and parameter sharing, showing that parameter sharing can become a stronger bottleneck than circuit topology itself. In contrast, [Prajapat et al. \(2025\)](#) demonstrate that a compact 45-parameter architecture can still achieve competitive binary-classification performance, whereas [Song et al. \(2024\)](#) reduce trainable parameters through shared QAOA filters while maintaining strong classification accuracy.

From an optimization perspective, a remarkable level of consistency exists across the literature. Cross-entropy is the dominant loss function and is used by most QCNN image-classification studies, including [Hur et al. \(2022\)](#), [Bokhan et al. \(2022\)](#), [Shi et al. \(2024\)](#), [Song et al. \(2024\)](#), and [Mordacci et al. \(2024\)](#). Notable exceptions include [Sun et al. \(2025\)](#), which employs mean squared error (MSE) together with gradient-descent backpropagation, and [Röseler et al. \(2025\)](#), which formulates classification as a regression-style task and optimizes a mean absolute error (MAE) objective. Similarly, Adam emerges as the de facto optimization algorithm across the majority of QCNN studies. In the most comprehensive optimizer comparison among the reviewed works, [Hur et al. \(2022\)](#) evaluate both Adam and Nesterov-based optimization and report significantly better performance with Adam. Subsequent QCNN architectures overwhelmingly adopt Adam as their default optimizer, suggesting an emerging consensus within the field.

Regarding evaluation methodology, classification accuracy is by far the most commonly reported metric and remains the primary criterion used to compare QCNN architectures. Nearly all image-classification studies report test or validation accuracy as their main performance indicator. Additional metrics are typically introduced only to provide complementary evidence. For example, [Mordacci et al. \(2024\)](#) report precision, recall, and F1-score for multiclass classification, while [Sun et al. \(2025\)](#) additionally evaluate recall, F1-score, and AUC. [Song et al. \(2024\)](#) emphasize ROC-based analysis, and [Liu and Lou \(2026\)](#) extend architectural evaluation through expressibility, entanglement capability, and gradient-based trainability metrics. Despite these additions, accuracy remains the dominant metric across the QCNN literature, making direct comparison possible even when architectures, datasets, and optimization procedures differ substantially.

2.3 Noise Robustness and NISQ Feasibility

Noise robustness is an increasingly important evaluation axis in QCNN research because most proposed architectures are intended for NISQ-era devices. However, the literature remains uneven in this regard as many studies report only noiseless simulation results, while fewer works explicitly evaluate the effect of quantum noise, measurement errors, or real-hardware execution. [Wei et al. \(2022\)](#) address this issue by simulating noisy circuits with stochastic Pauli errors, showing that their QCNN maintains similar performance under the tested noise model. [Song et al. \(2024\)](#) further analyze robustness to measurement errors in a resource-efficient QAOA-based QCNN, while [Liu and Lou \(2026\)](#) evaluate robustness under data noise and quantum noise channels, including bit-flip, phase-flip, and depolarizing noise.

Real-hardware validation remains comparatively scarce. Among the reviewed works, [Röseler et al. \(2025\)](#) is one of the few studies that reports QCNN execution on an IBM Heron r2 quantum processor, providing evidence beyond noiseless simulation. This distinction is important because high simulation accuracy does not necessarily imply hardware feasibility. Overall, noise robustness should be treated as a separate dimension of QCNN evaluation, alongside accuracy, parameter

efficiency, and qubit count. The current literature suggests that robust QCNN design requires not only expressive quantum filters, but also shallow circuits, efficient encodings, and architectures that limit accumulated gate and measurement errors.

3 Open Challenges and Research Opportunities

The literature reviewed in this survey reveals a field that is progressing rapidly but remains far from maturity. Current research is largely driven by three complementary objectives: the design of NISQ-compatible architectures with shallow circuits and limited parameter counts, the development of scalable QCNNs capable of processing increasingly complex visual data, and the search for trainable circuit ansätze that balance expressibility, entanglement capability, optimization stability, and robustness to noise. Despite significant architectural advances, the evidence remains fragmented across datasets, preprocessing pipelines, encoding strategies, qubit counts, and evaluation methodologies. Consequently, several fundamental challenges remain open.

- **Scalability beyond toy image-classification benchmarks remains unresolved.**

Most QCNN studies continue to rely on MNIST, Fashion-MNIST, binary classification tasks, reduced multiclass settings, or heavily compressed image representations. It remains unclear whether these approaches can scale to larger images, more classes, and more diverse datasets while maintaining their reported advantages. Future work should establish QCNN benchmarks involving higher-resolution images and more challenging visual tasks while explicitly accounting for state-preparation, circuit-execution, and measurement costs.

- **The role of feature encoding remains insufficiently understood.**

Encoding choices frequently influence QCNN performance as much as the convolutional architecture itself. The surveyed literature reports substantial performance differences arising from PCA, autoencoders, amplitude encoding, angle encoding, dense encoding, fragment encoding, and hybrid encoding schemes. Developing standardized encoding benchmarks and theoretical analyses of encoding-induced biases represents an important research direction.

- **Noise robustness remains an open challenge.**

Although QCNNs are frequently presented as NISQ-compatible architectures, relatively few studies systematically evaluate the impact of quantum noise. Existing results show that different architectures can respond very differently to the same noise process, suggesting that robustness is highly architecture-dependent. Future studies should incorporate hardware-calibrated noise models, error-mitigation techniques, and architecture-aware noise analysis across multiple datasets and quantum platforms.

- **Generalization ability is still insufficiently established.**

Several QCNN architectures achieve strong performance on reduced image-classification tasks, particularly in binary and low-class-count settings. The available evidence supports the feasibility of QCNNs on small-scale image-classification tasks, but their ability to generalize across datasets, domains, and increasing task complexity remains uncertain.

- **The lack of standardized benchmarks limits fair comparison.**

Current QCNN studies differ substantially in datasets, binary versus multiclass settings, preprocessing pipelines, encoding methods, ansatz families, optimizer settings, noise assump-

tions, qubit counts, and classical baselines. The absence of such a framework remains one of the most significant methodological limitations in the current QCNN literature.

- **Hardware deployment remains limited.**

Despite growing interest in practical quantum machine learning, most QCNN studies continue to rely on noiseless simulation or simulator-based noise models. Future work should prioritize hardware-aware evaluation, compare transpiled implementations across quantum platforms, and quantify the impact of state preparation, gate errors, and readout noise on end-to-end classification performance.

Overall, the literature suggests that the primary challenge facing QCNN research is no longer the construction of quantum convolutional architectures themselves, but rather the establishment of scalable, robust, and reproducible evaluation methodologies. Progress in encoding strategies, benchmarking standards, hardware validation, and noise-aware design will likely determine whether QCNNs evolve from promising research prototypes into practical tools for real-world image classification.

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